ROBUST AND UNCERTAINTY-AWARE VAE (RU-VAE) FOR ONE-CLASS CLASSIFICATION

Renuka Sharma[⋆†] *Suyash P. Awate*[⋆]

[⋆]Computer Science and Engineering Department, Indian Institute of Technology (IIT) Bombay, Mumbai.
[†]IITB-Monash Research Academy, Mumbai.

ABSTRACT

One-class classification (OCC) methods for abnormality detection learn either a generative model of the inlier class (e.g., using variants of kernel principal component analysis) or a decision boundary to encapsulate the inlier class (e.g., using one-class variants of the support vector machine). Recent methods use deep-neural-network models to learn (for the inlier class) either latent-space distributions or autoencoders, but *not* both. OCC learning typically relies solely on inlier-class data, but some recent semi-supervised versions also leverage some outlier-class training data. We propose a robust and uncertainty-aware variational framework for OCC, leveraging data-adaptive generalized-Gaussian (GG) models leading to distribution modeling in both latent space and image space. We propose a reparameterization for samples from the latent-space GG to enable backpropagation. Results on publicly available real-world datasets show the benefits of our method over others.

Index Terms— One-class classification, VAE, robustness, uncertainty, generalized Gaussian, semi-supervision.

1. INTRODUCTION AND RELATED WORK

Detecting abnormalities/outliers is a well-known one-class classification (OCC) problem, where completely characterizing the abnormal/outlier class using a finite-sized training sample is nearly infeasible because of the sample's inability to capture the large variability in the abnormal class. Hence, OCC learning relies on modeling the normal-class distribution or its associated envelope as the decision boundary.

Early methods for OCC use hand-crafted features within variants of density-based clustering [1], kernel principal component analysis (kPCA) [2], or one-class support vector machine [3]). Later methods use kernel mixture models [4]. Recent methods use deep neural networks (DNNs) to jointly learn a feature extractor and a classifier. DNN methods like DRAE [5] show better performance over kernel-based methods like [6]. Some methods [7, 5, 8, 9] learn an autoencoder for inliers, which reconstructs (i.e., encoding followed by decoding) inliers better than outliers. DeepSVDD [10] learns an encoder for normal-class images to map them to a compact subspace in the latent space, where the encoder maps abnormal-class images far from that subspace. AnoGAN [11] uses a generative adversarial DNN to learn a manifold for normal data and their variability within the manifold. DRAE [5] and RCAE [8] learn from training sets corrupted with outliers by weeding out outliers or reduce their effects. RCAE [8] uses inductive learning that extends robust PCA using an autoencoder. Unlike aforementioned methods, we propose variational learning by extending the VAE framework [12] to use data-adaptive generalized-Gaussian models subsuming robustness and uncertainty awareness.

OCC focuses mainly on learning from the normal class, but a small cohort of expert-labeled abnormal data can improve the estimation of the decision boundary enveloping the inlier-class latent-space distribution. Some non-DNN methods [13, 14, 15] leverage such limited information about the abnormal class using transductive learning. Typical DNN methods for OCC [7, 8, 10, 11] are unable to leverage such information, unlike the recently proposed SSAD [15], QSSAE [16], and DeepSAD [17]. QSSAE [16] extends DCAE to leverage such limited supervision; analogous to RCAE, QSSAE aims to be robust to the corruption in the training data. The semi-supervised OCC methods of SSAD [15] and DeepSAD [17] relate to SVDD that focuses on mapping normal data to a compact subspace within latent space; while SSAD extends SVDD, DeepSAD extends DeepSVDD. While SSAD and SVDD rely on hand-crafted features, hand-crafted kernels, and unsupervised learning frameworks, DeepSAD relies on DNN-based semi-supervised learning that leverages a small training set of expert-labeled anomalous images.

This paper proposes a novel variational formulation for OCC, i.e., robust and uncertainty-aware VAE (RU-VAE) that leverages data-adaptive generalized-Gaussian (GG) models leading to robust and uncertainty-aware distribution modeling in both latent space and image space. It proposes a reparameterization scheme for samples from the latent-space GG to enable optimization by backpropagation. We extend RU-VAE to leverage some outlier data during training to improve performance. Results on publicly available real-world datasets show the benefits of RU-VAE over existing methods.

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2. METHOD

We propose to learn a DNN-based one-class classifier to classify images into normal and abnormal classes. We propose to achieve our goal through a novel generalized variational-learning framework, RU-VAE (Robust and Uncertainty-aware VAE), that models both (i) the latent-space distribution conditioned on the input image (modeled using an encoder) as well as (ii) the image distribution conditioned on the latent-space encoding (modeled using a decoder) as generalized-Gaussian distributions that lead to robust fits. We also propose a reparameterization for the generalized-Gaussian to enable back-propagation based learning.

RU-VAE's encoder and decoder (Figure 1) output datum-specific GG distributions in latent space and image space. Section 2.1 proposes RU-VAE framework that leverages a small training set of labeled outliers as well. Section 2.2 proposes a latent-space GG reparameterization to enable back-propagation. In section 2.3, we present the abnormality score calculation strategy along with the techniques used for optimization of the RU-VAE framework.

2.1. RU-VAE's Statistical Model

Let R be a random field modeling a distribution on input medical images. Let Z be a *hidden/latent* random vector that captures all the information needed to generate image from R. Let a DNN-based *encoder* model a mapping $\Phi(\cdot;\theta^{\Phi})$, parameterized by θ^{Φ}, which maps the input image X to the parameters $\Phi(X;\theta^{\Phi})$ of a distribution on the latent vector Z. Let a DNN-based *decoder* model a mapping $\Psi(\cdot;\theta^{\Psi})$, parameterized by θ^{Ψ}, which maps the latent vector Z to the parameters $\Psi(Z;\theta^{\Psi})$ of a distribution on the image X. We propose to estimate the RU-VAE parameters $\theta^{\Phi\Psi} := \theta^{\Phi} \cup \theta^{\Psi}$ by maximizing the likelihood of the observed training-set images, i.e., images from the normal class $\{S_n\}_{n=1}^{N}$ and a small set of expert-labeled outlier images $\{T_m\}_{m=1}^{M}$.

Let $\mathcal{G}(\cdot;\mu,\alpha,\beta)$ be the factorized GG with mean vector μ, a vector of log-scale parameters α, and a vector of log-shape parameters β. Analogous to the VAE learning strategy [12], RU-VAE learning optimizes $\theta^{\Phi\Psi}$ for the underlying expectations. For each S_n, let z_{ni} be the i-th independent draw from the probability density function (PDF) $Q(Z|S_n;\theta^{\Phi}) = \mathcal{G}(Z;\Phi^{\mu}(S_n;\theta^{\Phi}),\Phi^{\alpha}(S_n;\theta^{\Phi}),\Phi^{\beta}(S_n;\theta^{\Phi}))$; the notation $z_{ni}(S_n,\theta^{\Phi})$ indicates that z_{ni} depends on encoder parameters θ^{Φ} and S_n. z_{ni} is reparameterized using the scheme in Section 2.2. Then, the learning formulation is equivalent to minimizing the loss

$$\mathcal{L}_1(\{S_n\}_{n=1}^{N};\theta^{\Phi\Psi}) := \frac{1}{NI}\sum_{n=1}^{N}\sum_{i=1}^{I}[0.5\|z_{ni}(S_n,\theta^{\Phi})\|_2^2 + \log\mathcal{G}(z_{ni}(S_n,\theta^{\Phi});\Phi^{\mu}(S_n;\theta^{\Phi}),\Phi^{\alpha}(S_n;\theta^{\Phi}),\Phi^{\beta}(S_n;\theta^{\Phi})) - \log\mathcal{G}(S_n;\Psi^{\mu}(Z_n;\theta^{\Psi}),\Psi^{\alpha}(Z_n;\theta^{\Psi}),\Psi^{\beta}(Z_n;\theta^{\Psi}))]. \quad (1)$$

For numerical stability and differentiability, we propose to evaluate $\log\mathcal{G}(a;\mu,\alpha,\beta)$ as follows. Let the operator $[\cdot]_d$ denote the d-th scalar component of its vector argument. Then, $[\log\mathcal{G}(b;\mu,\alpha,\beta)]_d := \log((\tau+\exp([\beta]_d))/(\Delta+\exp([\alpha]_d))) - (((b-[\mu]_d)/(\Delta+\exp([\alpha]_d)))^2+\epsilon)^{0.5(\tau+\exp([\beta]_d))} - \log(\Gamma(1/(\tau+\exp([\beta]_d)))) + \text{constant}$, where constants Δ, ϵ, and τ (all $\in \mathbb{R}^+$) act as regularizers.

Similar to $\mathcal{L}_1(\cdot)$, we propose a loss term for the expert-labeled outliers' set as $\mathcal{L}_2(\cdot)$. Because we want the outliers to be away from the compact subspace containing the inliers, we consider the negation of the terms (compared to $\mathcal{L}_1(\cdot)$) for latent-space and image-space for the outliers as follows.

$$\mathcal{L}_2(\{T_m\}_{m=1}^{M};\theta^{\Phi\Psi}) := -\frac{1}{MJ}\sum_{m=1}^{M}\sum_{j=1}^{J}[0.5\|z_{mj}(T_m,\theta^{\Phi})\|_2^2 + \log\mathcal{G}(z_{mj}(T_m,\theta^{\Phi});\Phi^{\mu}(T_m;\theta^{\Phi}),\Phi^{\alpha}(T_m;\theta^{\Phi}),\Phi^{\beta}(T_m;\theta^{\Phi})) - \log\mathcal{G}(T_m;\Psi^{\mu}(Z_m;\theta^{\Psi}),\Psi^{\alpha}(Z_m;\theta^{\Psi}),\Psi^{\beta}(Z_m;\theta^{\Psi}))]. \quad (2)$$

For each T_m, $z_{mj}(T_m,\theta^{\Phi})$ is the j-th independent draw from $Q(Z|T_m;\theta^{\Phi}) = \mathcal{G}(Z;\Phi^{\mu}(T_m;\theta^{\Phi}),\Phi^{\alpha}(T_m;\theta^{\Phi}),\Phi^{\beta}(T_m;\theta^{\Phi}))$, and is reparameterized using the scheme in Section 2.2.

To promote separability of the normal-class and abnormal-class distributions in latent space, we propose another loss term $\mathcal{L}_3(\{S_n\}_{n=1}^{N},\{T_m\}_{m=1}^{M};\theta^{\Phi\Psi})$. We introduce a nonlinear classifier $\mathcal{C}(\cdot;\theta^{\mathcal{C}})$ to discriminate between the distributions of the latent-space encodings of the independent draws $\cup_{n=1}^{N}\cup_{i=1}^{I}\{z_{ni}\}$ and $\cup_{m=1}^{M}\cup_{j=1}^{J}\{z_{mj}\}$, where z_{ni} and z_{mj} are independent draws from latent-space PDFs, as defined earlier, and reparameterized using the scheme in Section 2.2. Thus, the associated cross-entropy loss is

$$\mathcal{L}_3(\{S_n\}_{n=1}^{N},\{T_m\}_{m=1}^{M};\theta^{\Phi\Psi}) := -\frac{1}{MJ}\sum_{m=1}^{M}\sum_{j=1}^{J}\log(\mathcal{C}(z_{mj}(T_m,\theta^{\Phi});\theta^{\mathcal{C}})) - \frac{1}{NJ}\sum_{n=1}^{N}\sum_{j=1}^{J}\log(1-\mathcal{C}(z_{ni}(S_n,\theta^{\Phi});\theta^{\mathcal{C}})). \quad (3)$$

Thus, RU-VAE learning minimizes

$$\lambda_1\mathcal{L}_1(\{S_n\}_{n=1}^{N};\theta^{\Phi\Psi}) + \lambda_2\mathcal{L}_2(\{T_m\}_{m=1}^{M};\theta^{\Phi\Psi}) + \mathcal{L}_3(\{S_n\}_{n=1}^{N},\{T_m\}_{m=1}^{M};\theta^{\Phi\Psi}), \quad (4)$$

where $\lambda_1 > 0$ and $\lambda_2 > 0$ are free parameters.

2.2. Reparameterizing the latent vector for RU-VAE

To enable backpropagation-based optimization for encoder parameters θ^{Φ}, we propose to reparameterize the i-th independent draw $z_{ni} \sim \mathcal{G}(Z;\Phi^{\mu}(X_n;\theta^{\Phi}),\Phi^{\alpha}(X_n;\theta^{\Phi}),\Phi^{\beta}(X_n;\theta^{\Phi}))$ using a hierarchical (2-level) reparameterization scheme. Let

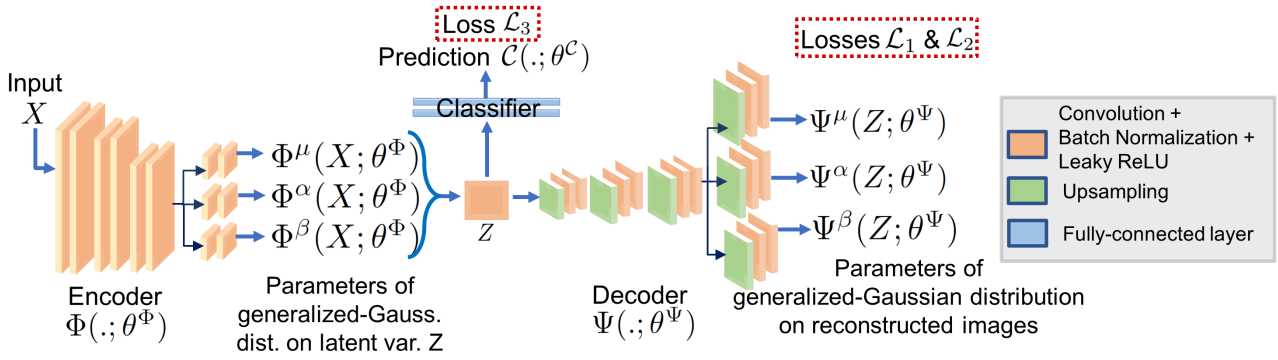


Fig. 1. **Robust and Uncertainty-aware VAE (RU-VAE) Framework for OCC.** The encoder has a common block before forking into 3 branches that output the mean $\Phi^{\mu}(\cdot;\theta^{\Phi})$, log-scale parameters $\Phi^{\alpha}(\cdot;\theta^{\Phi})$, and log-shape parameters $\Phi^{\beta}(\cdot;\theta^{\Phi})$. The common block has 3 sub-blocks, each using convolutions, pooling, batch normalization, and leaky-ReLU activation. Each branch has 1 sub-block with a fully-connected layer and leaky-ReLU activation. The decoder architecture is complementary to the encoder, as usual. The decoder's branches output the mean $\Psi^{\mu}(\cdot;\theta^{\Psi})$, log-scale parameters $\Psi^{\alpha}(\cdot;\theta^{\Psi})$, and log-shape parameters $\Psi^{\beta}(\cdot;\theta^{\Psi})$. The classifier has 2 fully-connected layers, with leaky-ReLU and sigmoid activations.

Gamma(a,b) be the Gamma PDF with shape parameter a and scale parameter b. First, we reparameterize [18] z_{ni} based on Gamma random variables as $[z_{ni}]_d := [\Phi^{\mu}(X_n;\theta^{\Phi})]_d + (\Delta + [\exp(\Phi^{\alpha}(X_n;\theta^{\Phi}))]_d)\, W_{nid}\, |Y_{nid}|^{1/(\tau+\exp([\Phi^{\beta}(X_n;\theta^{\Phi})]_d))}$, where random variables Y_{nid} have the PDF Gamma$(1/(\tau + [\Phi^{\beta}(S;\theta^{\Phi})]_d), 1)$ and W_{nid} take values ± 1 with probability 0.5. Second, we leverage implicit reparameterization gradients [19] for the Gamma random variables Y_{nid}.

2.3. Inference Strategy and Optimization Strategy

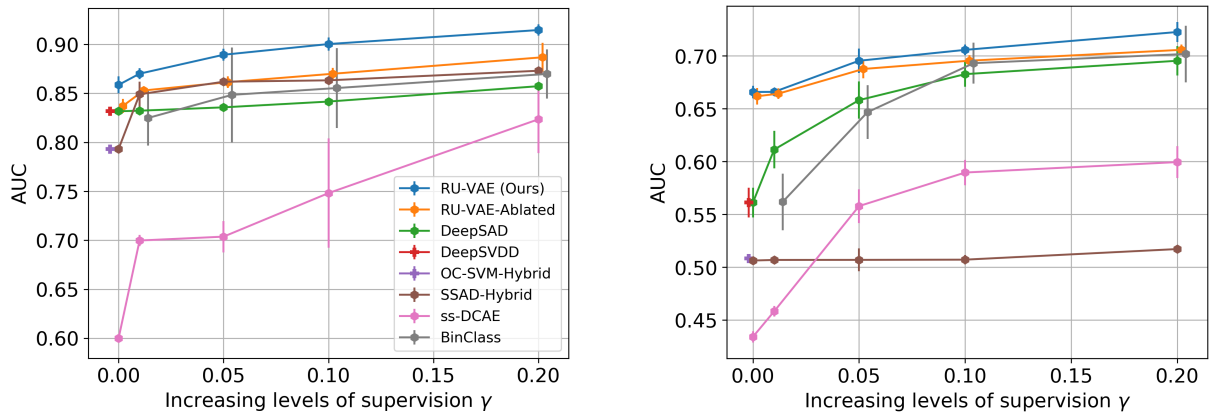
For inliers S, the loss $\mathcal{L}_1$ promotes the encoded mean $\Phi^{\mu}(S;\theta^{\Phi})$ to be close to the origin. Also, it promotes the decoder-output mean $\Psi^{\mu}(Z;\theta^{\Psi})$ to be close to the inlier S. During RU-VAE learning, for the outliers T, the objective function promotes (i) the encoded mean $\Phi^{\mu}(T;\theta^{\Phi})$ to be away from the origin, (ii) the latent-encoding distribution to be separated from that of inliers, and (iii) the decoder-output mean $\Psi^{\mu}(Z;\theta^{\Psi})$ to be far from the input T. In this paper, for a test image U, we propose an abnormality score as the norm of the mean $\Phi^{\mu}(U;\theta^{\Phi})$ (outliers lead to larger scores as compared to inliers), and then classify based on a threshold ρ (free parameter) on the score.

Our optimization is hierarchical. It first uses solely the inlier training set and excludes the branches modeling log-scale and log-shape (fixing GG scale parameters to 1 and shape parameters to 2). With this warm start, we include the log-scale and log-shape branches and retrain the DNN. Last, we include the outlier training set and the classifier branch and retrain the DNN.

3. RESULTS AND DISCUSSION

We compare with "semi-supervised" methods that learn using both $\{S_n\}_{n=1}^N$ and $\{T_m\}_{m=1}^M$, and "unsupervised" methods that learn using $\{S_n\}_{n=1}^N$. The 6 **baseline methods** are: (i)-(ii) DeepSAD [17] (semi-supervised DNN-based OCC) and DeepSVDD [10] (unsupervised DNN-based OCC); (iii) ss-DCAE: a semi-supervised version of the DNN-based OCC in [7] relying on image reconstruction using an autoencoding strategy; (iv)-(v) SSAD-Hybrid [20] (semi-supervised kernel-based OCC extending SSAD [15]) and OC-SVM-Hybrid [21, 20] (unsupervised kernel-based OCC extending OC-SVM [22]) using pre-extracted autoencoder-based features and the radial-basis-function kernel; (vi) BinClass: DNN-based binary classifier using $\{S_n\}_{n=1}^N$ and $\{T_m\}_{m=1}^M$ when $M > 0$. We also compare with RU-VAE-Ablated, i.e., an **ablated** version of RU-VAE without the classifier. We evaluate at different levels of supervision $\gamma := M/(M+N)$ ranging within $[0, 0.2]$. Quantitative evaluation uses the area under the receiver-operating-characteristics curve (AUC).

We use 2 publicly available **datasets**: (i) the Broad Bioimage Benchmark Collection (`bbbc.broadinstitute.org/BBBC041`) has 1328 microscopy images of blood, each containing red blood cells manually segmented and labeled as malaria-infected (abnormal) or not (normal); we classify patches of size 170×170 pixels (cell diameter is typically 170 pixels); we randomly extract 10000 normal patches and 2500 abnormal patches; (ii) the Nanofiber dataset (`www.mi.imati.cnr.it/ettore/NanoTWICE/`) has 45 electron-microscopy images of nanofiber biomaterials, with 40 images showing material defects (manually segmented); analogous to the malaria dataset, we randomly extract 17228 normal patches and 5452 abnormal patches of size 128×128 pixels. We consider a patch as abnormal if at least 50% of its pixels are labeled abnormal. All methods resize patches to 32×32 pixels. We partition the datasets into 3 subsets for training, validation (to tune free parameters), and testing (test set contains an equal number of normal and abnormal patches). Infected RBCs are of 6 types, with varying



(a) Blood Microscopy in Malaria

(b) Biomaterial (Nanofibers) Microscopy

Fig. 2. **Results on Malaria and Nanofiber Datasets.** AUC values for RU-VAE and baseline methods. The plots (with bars) indicate the variability in AUC across randomly sampled training sets, validation sets, and test sets (20 repeats).

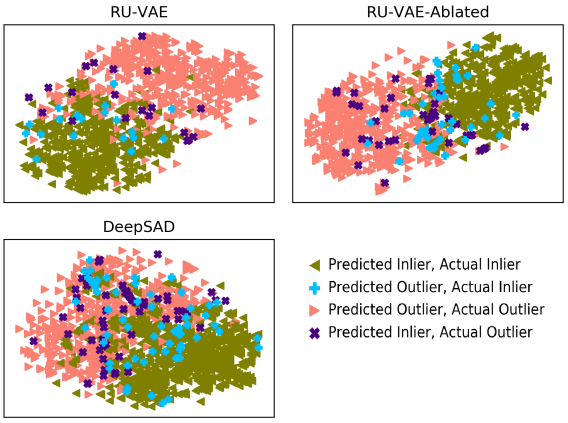


Fig. 3. t-SNE plots of latent-space encodings for Malaria data.

natural frequency of occurrence. To realize a OCC scenario, training uses abnormal patches from only 3 infected-RBC types, and testing uses patches from all 6 infected-RBC types. As motivated in the DeepSAD paper [17], we pollute/corrupt the training set by misclassifying 10% of outliers as inliers to evaluate the robustness of the learned models.

Results. With increasing levels of semi-supervision γ, DeepSAD improves over DeepSVDD, and SSAD-Hybrid improves over OC-SVM-Hybrid. For these datasets, DeepSAD and DeepSVDD that are solely based on latent-space-loss modeling improve over ss-DCAE that is solely based on image-space-loss modeling. In contrast, RU-VAE that combines robust GG-based models in latent space and image space improves over both DeepSAD and ss-DCAE. SSAD-Hybrid, which employs fixed DNN-based features, significantly underperforms on the Nanofiber dataset. RU-VAE outperforms other methods at all levels of semi-supervision $\gamma \geq 0$ (Figure 2). RU-VAE improves over RU-VAE-Ablated that is devoid of the classifier that aids RU-VAE in separating the inlier and outlier distributions in latent space. Even RU-VAE-Ablated, using variational inference with robust GG models in both latent and image space, compares favorably to other DNN-based methods at all levels of semi-supervision $\gamma \geq 0$. BinClass's performance drops more drastically at low levels of semi-supervision γ because of the paucity of the outlier-class data; indeed, BinClass cannot perform at $\gamma = 0$. The t-SNE [23] plots of latent-space encodings (Figure 3) show improved separability between the normal and abnormal classes for RU-VAE, compared to DeepSAD and RU-VAE-Ablated, at $\gamma = 0.2$ level of supervision.

4. CONCLUSION

This paper proposes RU-VAE that is a novel generalization of the VAE framework for OCC that also leverages a small training set of labeled outliers. RU-VAE leverages factored GG models for, both, robust heavy-tailed modeling (through GG shape parameters) and uncertainty-aware heteroscedastic modeling (through GG scale parameters) in latent space and image space; the GG models adapt to the specific input datum. We propose a reparameterization for sampling from the latent-space GG to enable backpropagation. Comprehensive empirical analyses, involving 6 baselines and an ablated version on two public datasets, show the benefits of our method over the state of the art and provide insights into the benefits of the novel components.

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Conflicts of Interest

The authors have no conflicts of interest to disclose.

Compliance with Ethical Standards

This research study was conducted retrospectively using human subject data made available in open access. Ethical approval was not required as confirmed by the license attached with the open access data.